Statistics Hybrid Practice Set – S14 (L10)

New Focus: Five Number Summary + Box Plot

Topics Covered

- Mean, Median, Mode
- Range
- Variance
- Standard Deviation
- Percentiles
- Quartiles
- **Five Number Summary — New** ⭐
- **Box Plot — New** ⭐
- IQR
- Outlier Detection
- Histogram interpretation
- Frequency Distribution

Instructions

- Solve on **paper first**.
- Then verify in **Excel**.
- Q1 is intentionally **not** Population/Sample identification.
- No formula-recall question.

Q1. Mean vs Median — Business Interpretation

A company recorded the following customer spending amounts:

₹420, ₹450, ₹480, ₹500, ₹520, ₹550, ₹2,000

Answer:

1. Calculate the **Mean**.
2. Calculate the **Median**.
3. Which measure better represents typical customer spending?
4. Why?

Excel

Verify your calculations.

Q2. Quartile Calculation

The following values represent delivery times in minutes:

12, 15, 18, 20, 22, 24, 26, 28, 30

Find:

1. Q1
2. Q2
3. Q3
4. IQR

Excel

Verify your answers using QUARTILE.INC().

⭐ Q3. Five Number Summary — New

Using the same dataset from Q2:

12, 15, 18, 20, 22, 24, 26, 28, 30

Find the **Five Number Summary**:

1. Minimum
2. Q1
3. Median
4. Q3
5. Maximum

Important

The **Five Number Summary** is:

Minimum → Q1 → Median → Q3 → Maximum

⭐ Q4. Interpret a Five Number Summary

A company's employee salary dataset has this Five Number Summary:

Statistic	Value
Minimum	₹18,000
Q1	₹25,000
Median	₹32,000
Q3	₹45,000
Maximum	₹90,000

Answer:

1. What is the median salary?
2. Approximately what percentage of employees earn **₹45,000 or less**?
3. What is the IQR?
4. Which part of the distribution shows the largest spread: **Q1 → Median** or **Median → Q3**?

Explain your answer.

⭐ **Q5. Build a Box Plot**

Use this dataset:

10, 12, 14, 15, 16, 18, 20, 22, 24, 26, 28

Paper

Identify:

- Minimum
- Q1
- Median
- Q3
- Maximum

Then sketch a **box plot** manually.

Excel

Create a **Box & Whisker chart** using the dataset.

Q6. Box Plot Interpretation

A box plot has the following values:

- Minimum = 20
- Q1 = 30
- Median = 40
- Q3 = 55
- Maximum = 80

Answer:

1. What is the IQR?
2. What percentage of observations lie between Q1 and Q3?
3. Is the median closer to Q1 or Q3?
4. What does this tell you about the distribution?

Q7. Outlier Detection + Box Plot

Dataset:

15, 16, 18, 19, 20, 21, 22, 24, 26, 70

Find:

1. Q1
2. Q3
3. IQR
4. Lower Fence
5. Upper Fence
6. Is 70 an outlier?

Excel

Verify all calculations.

Interpretation

If 70 is an outlier, explain how it would appear on a **box plot**.

Q8. Compare Two Datasets

Two teams have these delivery times:

Team A

18, 19, 20, 21, 22, 23, 24

Team B

10, 15, 20, 21, 22, 30, 40

Answer:

1. Which team has the greater range?
2. Which team appears more spread out?
3. Which team's box plot would likely have longer whiskers?
4. Which team appears more consistent?

Give a short reason for each answer.

Q9. Real Data Analyst Scenario

A Data Analyst wants to show:

- Minimum
- Q1
- Median
- Q3
- Maximum
- Possible outliers

in **one visual**.

Which visualization is most appropriate?

A. Histogram
B. Box Plot
C. Pie Chart
D. Line Chart

Explain why.

Q10. Mixed Business Case

An online store has the following order values:

₹300, ₹320, ₹340, ₹350, ₹360, ₹380, ₹400, ₹420, ₹450, ₹1,500

Answer:

1. Which measure would better represent the typical order value: **Mean or Median**?
2. Is there likely an outlier?
3. Which method would you use to confirm it?
4. Which visualization would be useful for showing the distribution and outlier?
5. Explain why the chosen visualization is useful.